Problems on Exact Differential Equation.

1. Solve $(\tan y + x)\,dx + (x\sec^2 y - 3y)\,dy = 0$

Soln: Let $(\tan y + x)\,dx + (x\sec^2 y - 3y)\,dy = 0$ — ①

we have, $M = \tan y + x$

$N = x\sec^2 y - 3y$

$$\frac{\partial M}{\partial y} = \sec^2 y \quad \text{and} \quad \frac{\partial N}{\partial x} = \sec^2 y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

∴ The given differential equation ① is an exact Differential equation.

The general solution of Equation ① is given as

$$\int M\,dx + \int [\text{Terms in } N \text{ free from } x]\,dy = C$$

y = constant

$$\int (\tan y + x)\,dx - 3\int y\,dy = C$$

$$\therefore x\tan y + \frac{x^2}{2} - \frac{3y^2}{2} = C$$

2. Solve $(1 + \log xy)\,dx + \left(1 + \frac{x}{y}\right)dy = 0$

Soln: Let $(1 + \log xy)\,dx + \left(1 + \frac{x}{y}\right)dy = 0$ — ①

be the given Differential equation.

Now comparing with $Mdx + Ndy = C$ we get

$M = 1 + \log xy$

$N = 1 + \frac{x}{y}$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(1 + \log xy)$$

$$= \frac{\partial 1}{\partial y} + \frac{\partial}{\partial y}(\log xy) = 0 + \frac{1}{xy}\frac{\partial (xy)}{\partial y}$$

$$\frac{\partial M}{\partial y} = \frac{1}{xy}\left(x\frac{\partial y}{\partial y}\right) = \frac{x}{xy} = \frac{1}{y}$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left(1 + \frac{x}{y}\right) = \frac{\partial 1}{\partial x} + \frac{\partial}{\partial x}\left(\frac{x}{y}\right)$$

$$\frac{\partial N}{\partial x} = 0 + \frac{1}{y}\frac{\partial x}{\partial x} = \frac{1}{y}$$

Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ The given Differential equation is exact

The general solution of equation ① is given as

$$\int_{y=\text{constant}} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = C$$

$$\therefore \int_{y=\text{constant}} (1+\log xy)\,dx + \int (1)\,dy = C$$

$$\therefore \int dx + \int \log xy\,dx + \int dy = c$$

$$\therefore x + \left[\log xy \int dx - \int \left(\frac{d}{dx}\log xy \int dx\right) dx\right] + y = C \quad \text{— [By applying integrating by parts]}$$

$$\therefore x + \log xy\,(x) - \int \left[\left(\frac{1}{xy}\frac{d(xy)}{dx}\right)(x)\right] dx + y = C$$

$$\therefore x + x\log xy - \int \left(\frac{1}{xy} \times y \times x\right) dx + y = C$$

$$\therefore x + x\log xy - \int dx + y = C$$

$$\therefore x + x\log xy - x + y = C$$

$$\therefore x\log xy + y = 0$$

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3. Solve $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

Soln: Let, $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

$(\sin x + x\cos y + x)\,dy = -(y\cos x + \sin y + y)\,dx$

$(y\cos x + \sin y + y)\,dx + (\sin x + x\cos y + x)\,dy = 0 \quad -①$

we have,

$M = y\cos x + \sin y + y$ and $N = \sin x + x\cos y + x$

Consider,

$\frac{\partial M}{\partial y} = \cos x + \cos y + 1$ and $\frac{\partial N}{\partial x} = \cos x + \cos y + 1$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ The given DE is exact.

The general solution of equation ① is given as $\int M\,dx + \int [\text{Terms in } N \text{ free from } x]\,dy = C$

$\int (y\cos x + \sin y + y)\,dx = C$

$\therefore y\sin x + x\sin y + xy = C$

4. Solve $\left[y\left(1+\frac{1}{x}\right) + \cos y\right]dx + [x + \log x - x\sin y]\,dx = 0$

Soln:- Let $\left[y\left(1+\frac{1}{x}\right) + \cos y\right]dx + [x + \log x - x\sin y]\,dx = 0 \quad -①$

be the given equation.

Comparing given differential equation with $M\,dx + N\,dy = 0$, we get

$M = y\left(1+\frac{1}{x}\right) + \cos y$ and $N = x + \log x - x\sin y$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left[y\left(1+\frac{1}{x}\right)+\cos y\right]$$

$$= \left(1+\frac{1}{x}\right)\frac{\partial y}{\partial y} + \frac{\partial \cos y}{\partial y}$$

$$= \left(1+\frac{1}{x}\right)+(-\sin y)$$

$$= 1+\frac{1}{x} - \sin y$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}\left[x+\log x - x\sin y\right]$$

$$= \frac{\partial}{\partial x} + \frac{\partial \log x}{\partial x} - \sin y \frac{dx}{\partial x}$$

$$= 1+\frac{1}{x} - \sin y$$

Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ The given differential equation (1) is an exact DE.

The general solution of equation (1) is given as

$$\int M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = C$$

y = constant

$$\therefore \int_{y=\text{constant}} \left[y\left(1+\frac{1}{x}\right)+\cos y\right]dx + \int (0)\,dy = C$$

$$\therefore y\int\left(1+\frac{1}{x}\right)dx + \cos y\int dx = C$$

$$\therefore y[x+\log x] + \cos y\,[x] = C$$

$$[x+\log x]y + x\cos y = C$$

Alternatively,

General solution can also be given as

$$\int_{x=\text{constant}} N\,dy + \int (\text{terms of } M \text{ not containing } y)\,dx = C$$

$$\therefore \int (x+\log x - x\sin y)\,dy + \int (0)\,dx = C$$

$$\therefore x\int dy + \log x \int dy - x\int \sin y\,dy = C$$

$$\therefore x(y) + \log x\,(y) - x\,(-\cos y) = C$$

$$\therefore \boxed{xy + y\log x + x\cos y = C}$$

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5. Solve $\frac{dy}{dx} = \frac{y+1}{(y+2)e^y - x}$

Soln: Let $\frac{dy}{dx} = \frac{y+1}{(y+2)e^y - x}$ be the given DE.

$(y+1)\,dx - [(y+2)e^y - x]\,dy = 0$ — ①

Now comparing with $M\,dx + N\,dy = 0$, we get

$M = y+1$

$N = -[(y+2)e^y - x] = -(y+2)e^y + x$

$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(y+1) = \frac{\partial y}{\partial y} + \frac{\partial 1}{\partial y} = 1 + 0 = 1$

$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}[-(y+2)e^y + x]$

$= -(y+2)e^y \frac{\partial 1}{\partial x} + \frac{\partial x}{\partial x}$

$= -(y+2)e^y (0) + 1 = 1$

Here, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ The given DE is exact.

∴ The general solution of Equation ① given as

$\int_{y=\text{constant}} M\,dx + \int [\text{Terms of } N \text{ not containing } x]\,dy = C$

$\therefore \int_{y=\text{constant}} (y+1)\,dx + \int [-(y+2)e^y]\,dy = C$

$\therefore y\int dx + \int dx - \int (y+2)e^y\,dy = C$

$\therefore y(x) + (x) - [e^y(y+2) - e^y(1+0)] = C$

[∵ Using integrating by parts]

$\therefore xy + x - (y+2-1)e^y = C$

$\therefore xy + x - (y+1)e^y = C$

6. Solve $\frac{dy}{dx} = \frac{\tan y - 2xy - y}{x^2 - x\tan^2 y + \sec^2 y}$

Soln:- Let $\frac{dy}{dx} = \frac{\tan y - 2xy - y}{x^2 - x\tan^2 y + \sec^2 y}$ be the given DE.

$(\tan y - 2xy - y)\,dx - (x^2 - x\tan^2 y + \sec^2 y)\,dy = 0$ — ①

Now equating with $M\,dx + N\,dy = 0$, we get

$M = \tan y - 2xy - y$

$N = -(x^2 - x\tan^2 y + \sec^2 y)$

$= -(x^2 + x\tan^2 y - \sec^2 y)$

$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(\tan y - 2xy - y)$

$= \frac{\partial}{\partial y}(\tan y) - 2x\frac{\partial y}{\partial x} - \frac{\partial y}{\partial y}$

$= \sec^2 y - 2x - 1$

$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(-x^2 + x\tan^2 y - \sec^2 y)$

$= \frac{\partial}{\partial x}(-x^2) + \tan^2 y\frac{\partial x}{\partial x} - \sec^2 y\frac{\partial 1}{\partial x}$

$= -2x + \tan^2 y(1) - \sec^2 y(0)$

$= -2x + \tan^2 y$

$= -2x + \sec^2 y - 1 \quad [\because \tan^2 y = \sec^2 y - 1]$

Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

The given differential equation ① is an Exact equation

The general solution of equation ① is given as

$\int M\,dx + \int$(terms of N not containing x) $dy = C$ (y = constant)

$\therefore \int(\tan y - 2xy - y)\,dx + \int(-\sec^2 y)\,dy = C$ (y = constant)

$\therefore \tan y\int dx - 2y\int x\,dx - y\int dx - \int \sec^2 y\,dy = C$

$\therefore \tan y(x) - 2y\left(\frac{x^2}{2}\right) - y(x) - \tan y = C$

$\therefore \boxed{x\tan y - x^2 y - xy - \tan y = C}$

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Problems on Rule 1:

71. Solve $(2x\log x - xy)dy + 2y\,dx = 0$

Soln:- Let $(2x\log x - xy)dy + 2\,dx = 0$ — ①

Compairing with $M\,dx + N\,dy = 0$, we get

$M = 2y$

$N = 2x\log x - xy$

$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(2y) = \frac{\partial y}{\partial y} = (2 \times 1) = 2$

$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(2x\log x - xy)$

$= 2\frac{\partial(x\log x)}{\partial x} - y\frac{\partial x}{\partial x}$

$= 2\left[\frac{\partial x}{\partial x}\log x + x\frac{\partial \log x}{\partial x}\right] - y$

$= 2[\log x + x(1/x)] - y$

$= 2\log x + 2 - y$

Here $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

The given DE ① is not an exact DE.

$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2 - (2\log x + 2 - y)$

$= 2 - 2\log x - 2 + y$

$= -2\log x + y$

$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-2\log x + y}{2x\log x - xy} = \frac{-(2\log x - y)}{x(2\log x - y)} = -\frac{1}{x}$

$= f(x)$

$IF = e^{\int f(x)\,dx} = e^{\int -\frac{1}{x}\,dx} = e^{-\int \frac{1}{x}\,dx}$

$= e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$

Multiplying given differential equation by an IF

$[(2x\log x - xy)dy]\frac{1}{x} + (2y\,dx)\frac{1}{x} = 0$

$(2x\log x - y)dy + \frac{2y}{x}dx = 0$ — (2)

$M' = \frac{2y}{x}$ and $N' = 2\log x - y$

$$\frac{\partial M'}{\partial y} = \frac{\partial}{\partial y}\left(\frac{2y}{x}\right) = \frac{2\,\partial y}{x\,\partial y} = \frac{2}{x}(1) = \frac{2}{x}$$

$$\frac{\partial N'}{\partial x} = \frac{\partial}{\partial x}(2\log x - y) = 2\frac{\partial \log x}{\partial x} - \frac{\partial 1}{\partial y}$$

$$= 2\left(\frac{1}{x}\right) - y(0) = \frac{2}{x}$$

Here, $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$

So Equation (2) is an exact equation.

The general solution of Equation is given as,

$\int M'dx + \int$ (terms of N' not containing x) $dy = C$
y = constant

$$\therefore \int_{y=\text{constant}} \frac{2y}{x}dx + \int(-y)dy = C$$

$$\therefore 2y\int\frac{1}{x}dx - \int y\,dy = C$$

$$\therefore 2y\log x - \frac{y^2}{2} = C$$

2. Solve $(xy^2 - e^{1/x^3})dx - x^2y\,dy = 0$

Soln:- Let $(xy^2 - e^{1/x^3})dx - x^2y\,dy = 0$ — (1) be the Differential Equation.

we have,

$M = xy^2 - e^{1/x^3}$ and $N = -x^2y$

we get,

$\frac{\partial M}{\partial y} = 2xy$ and $\frac{\partial N}{\partial x} = -2xy$

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Consider, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2xy - (-2xy)}{-x^2y} = \frac{4xy}{-x^2y}$

$= \frac{-4}{x} = f(x)$

$\therefore \int f(x)\,dx = \int \frac{-4}{x}\,dx = \log\left(\frac{1}{x^4}\right)$

$\therefore IF = e^{\int f(x)\,dx} = e^{\log(1/x^4)} = \frac{1}{x^4}$

$IF = \frac{1}{x^4}$

Multiplying equation (i) by Integrating factor, we get

$\frac{1}{x^4}(xy^2 - e^{1/x^3})\,dx - \frac{1}{x^4}(x^2y)\,dy = 0$

$\left(\frac{y^2}{3} - \frac{e^{1/x^3}}{x^4}\right)dx - \frac{y}{x^2}\,dy = 0$ — (ii)

$\left\{\frac{2y}{x^3}, \frac{2y}{x^3}\right\}$

The Differential Equation (ii) is exact

The general solution of eqn (ii) is given as

$\int M\,dx + \int [\text{Terms in } N \text{ free from } x]\,dy = c$

$y = \text{constant}$

$\int\left(\frac{y^2}{x^3} - \frac{e^{1/x^3}}{x^4}\right)dx = c$

$\int \frac{y^2}{x^3}\,dx - \int \frac{e^{1/x^3}\,dx}{x^4}$

$\frac{y^2}{-2x^2} - \left(\frac{e^{1/x^3}}{-3}\right) = c$

$\therefore \frac{e^{1/x^3}}{3} - \frac{y^2}{2x^2} = c$

3. Solve $x \sin x\, dy + (xy \cos x - y \sin x - 2)\, dx = 0$

Soln:- Let $x \sin x\, dy + (xy \cos x - y \sin x - 2)\, dx = 0$ — (1) be the given DE.

Compairing with $M dx + N dy = 0$, we get

$M = x^2 + y^2 + 1$

$N = -2xy$

$$\frac{\partial M}{\partial N} = \frac{\partial (x^2+y^2+1)}{\partial y} = \frac{\partial x^2}{\partial y} + \frac{\partial y^2}{\partial y} + \frac{\partial 1}{\partial y}$$

$$= 0 + 2y + 0 = 2y$$

$$\frac{\partial N}{\partial x} = \frac{\partial (-2xy)}{\partial x} = -2y \frac{\partial x}{\partial x} = -2y(1) = -2y$$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$.

Given differential equation is not an exact equation.

Consider,

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2y - (-2y) = 4y$$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4y}{-2xy} = -\frac{2}{x} = f(x)$$

Now we have $IF = e^{\int f(x) dx} = e^{\int -\frac{2}{x} dx} = e^{-2\int \frac{1}{x} dx}$

$= e^{-2(\log x)} = e^{\log x^{-2}} = x^{-2} = \frac{1}{x^{-2}}$

Multiplying given differential Equation (1) by IF, we get

$$[(x^2+y^2+1)dx]\frac{1}{x^2} - (2xy\, dy)\frac{1}{x^2} = 0$$

$$\left(\frac{x^2+y^2+1}{x^2}\right) dx - \left(\frac{2xy}{x^2}\right) dy = 0 \quad \text{— (2)}$$

Consider, $M' = 1 + \frac{y^2}{x^2} + \frac{1}{x^2}$

$N' = -\frac{2y}{x}$

$$\frac{\partial M'}{\partial y} = \frac{\partial}{\partial y}\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right) = \frac{\partial 1}{\partial y} + \frac{1}{x^2}\frac{\partial y^2}{\partial y} + \frac{1}{x^2}\frac{\partial 1}{\partial y}$$

$$= \frac{2y}{x^2}$$

$$\frac{\partial N'}{\partial x} = \frac{\partial}{\partial x}\left(\frac{-2y}{x}\right) = -2y\frac{\partial \frac{1}{x}}{\partial x}$$

$$= -2y\left(\frac{-1}{x^2}\right) = \frac{2y}{x^2}$$

Now $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$

So Equation ② is an Exact Differential Equation

The general solution of Equation ② is given as

$$\int_{y=\text{constant}} M'dx + \int(\text{terms of } N' \text{ not containing } x)dy = C$$

$$\therefore \int\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx + \int(0)\,dy = C$$

$$\therefore \int dx + y^2\int\frac{1}{x^2}dx + \int\frac{1}{x^2}dx = C$$

$$\therefore x + y^2\left[\frac{x^{-2}+1}{-2+1}\right] + \left[\frac{x^{-2}+1}{-2+1}\right] = C$$

$$\therefore x - \frac{y^2}{x} - \frac{1}{x} = C$$

4. Solve $(x^3e^x - my^2)dx + mxydx = 0$

Soln: Let $(x^3e^x - my^2)dx + mxydx = 0$ — ①

be the given DE.

Now comparing with $Mdx + Ndy = 0$, we get

$M = x^3e^x - my^2$ and $N = mxy$

$\frac{\partial M}{\partial y} = -2my$ and $\frac{\partial N}{\partial x} = my$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ The given DE (1) is not exact

Now we consider,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-2my - my}{mxy} = \frac{-3my}{mxy} = -\frac{3}{x} = f(x)$$

We have $IF = e^{\int f(x)dx} = e^{\int -3/x\, dx} = e^{-3\log x} = x^{-3} = \frac{1}{x^3}$

Multiplying Differential equation (1) by IF,

$$(x^3e^x - my^2)dx \times \frac{1}{x^3} + mxy\,dy \times \frac{1}{x^3} = 0$$

$$\left(e^x - \frac{my^2}{x^3}\right)dx + \frac{mxy}{x^2}dy = 0$$

Now, $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x} = -\frac{2my}{x^3}$

$\therefore$ Integrating M' with respect to x:

$$\int \left(e^x - \frac{my^2}{x^3}\right)dx = e^x + \frac{my^2}{2x^2} + C$$

Integrating N' with respect to x:

$$\int 0\,dx = C$$

$$\therefore e^x + \frac{my^2}{2x^2} = C$$

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Problems on Rule 2

1. Solve $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$

Soln: Let $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$ — (1) be the given DE.

Comparing with $Mdx + Ndy = 0$, we get

$M = 2xy^4e^y + 2xy^3 + y$ and $N = x^2y^4e^y - x^2y^2 - 3x$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(2xy^4e^y + 2xy^3 + y)$$

$$= 2x\frac{\partial(y^4e^y)}{\partial y} + 2x\frac{\partial y^3}{\partial y} + \frac{\partial y}{\partial y}$$

$$= 2x\left[\frac{\partial y^4}{\partial y}e^y + y^4\frac{\partial e^y}{\partial y}\right] + 2x(3y^2) + 1$$

$$= 2x[(4y^3)e^y + y^4e^y] + 6xy^2 + 1$$

$$= 8xy^3e^y + 2xy^4e^y + 6xy^2 + 1$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x^2y^4e^y - x^2y^2 - 3x)$$

$$= y^4e^y\frac{\partial x^2}{\partial x} - y^2\frac{\partial x^2}{\partial x} - 3\frac{\partial x}{\partial x}$$

$$= y^4e^y(2x) - y^2(2x) - 3(1)$$

$$= 2xy^4e^y - 2xy^2 - 3$$

Here $\frac{\partial N}{\partial x} \neq \frac{\partial M}{\partial y}$

The given DE (1) is not exact DE.

Consider,

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = (2xy^4e^y - 2xy^2 - 3) - (8xy^3e^y + 2xy^4e^y + 6xy^2 + 1)$$

$$= 2xy^4e^y - 2xy^2 - 3 - 8xy^3e^y - 2xy^4e^y - 6xy^2 - 1$$

$$= -8xy^3e^y - 8xy^2 - 4$$

we have,

$$\frac{\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}}{M} = \frac{-8xy^3e^y-8xy^2-4}{2xy^4e^y+2xy^3+y}$$

$$= \frac{-4(2xy^3e^y+2xy^2+1)}{y(2xy^3e^y+2xy^2+1)} = -\frac{4}{y} = g(y)$$

Now we have, $IF = e^{\int g(y)\,dy} = e^{\int -\frac{4}{y}\,dy} = e^{-4\int \frac{1}{y}\,dy}$

$= e^{-4\log y} = e^{\log y^{-4}} = y^{-4} = \frac{1}{y^4}$

Multiplying Equation ① by IF, we get,

$$[(2xy^4e^y+2xy^3+y)\,dx]\frac{1}{y^4} + [(x^2y^4e^y-x^2y^2-3x)\,dy]\frac{1}{y^4} = 0$$

$$\left(2xe^y+\frac{2x}{y}+\frac{1}{y^3}\right)dx + \left(x^2e^y-\frac{x^2}{y^2}-\frac{3x}{y^4}\right)dy = 0 \quad \text{—— ②}$$

$M' = 2xe^y+\frac{2x}{y}+\frac{1}{y^3}$

$N' = x^2e^y-\frac{x^2}{y^2}-\frac{3x}{y^4}$

$$\frac{\partial M'}{\partial y} = \frac{\partial}{\partial y}\left(2xe^y+\frac{2x}{y}+\frac{1}{y^3}\right)$$

$$= 2x\frac{\partial e^y}{\partial y} + 2x\frac{\partial(1/y)}{y} + \frac{\partial(1/y^3)}{\partial y}$$

$$= 2x(e^y) + 2x\left(-\frac{1}{y^2}\right) + \left(-\frac{3}{y^4}\right)$$

$$= 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}$$

$$\frac{\partial N'}{\partial x} = \frac{\partial}{\partial x}\left(x^2e^y-\frac{x^2}{y^2}-\frac{3x}{y^4}\right)$$

$$= e^y\frac{\partial x^2}{\partial x} - \frac{1}{y^2}\frac{\partial x^2}{\partial x} - \frac{3\partial x}{y^4\partial x}$$

$$= e^y(2x) - \frac{1}{y^2}(2x) - \frac{3}{y^4} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}$$

Now, $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$

∴ The Equation (2) is an exact differential Equation

∴ The general solution of Equation (2) is given as,

$$\int_{y=\text{constant}} M'\,dx + \int (\text{terms of } N' \text{ not containing } x)\,dy = C$$

$$\therefore \int \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right) dx + \int 0\,dy = C$$

$$\therefore 2e^y \int x\,dx + \frac{2}{y}\int x\,dx + \frac{1}{y^3}\int dx = C$$

$$\therefore 2e^y\left(\frac{x^2}{2}\right) + \frac{2}{y}\left(\frac{x^2}{2}\right) + \frac{1}{y^3}(x) = C$$

$$\therefore x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = C$$

2. Solve $\left(\frac{y}{x}\sec y - \tan y\right)dx - (x - \sec y \log x)\,dy = 0$

Soln:- Let $\left(\frac{y}{x}\sec y - \tan y\right)dx - (x - \sec y \log x)\,dy = 0$ — (1)

be the given equation.

Comparing $M\,dx + N\,dy = 0$, we get

$$M = \frac{y}{x}\sec y - \tan y$$

$$N = -(x - \sec y \log x)$$

$$\frac{\partial M}{\partial y} = \frac{1}{x}\sec y + \frac{y}{x}\sec y \tan y - \sec^2 y$$

$$\frac{\partial N}{\partial x} = -\left(1 - \frac{1}{x}\sec y\right)$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

The given equation is not an exact differential equation.

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

$$= \frac{-(1-\frac{1}{x}\sec y) - (\frac{1}{x}\sec y + \frac{y}{x}\sec y \tan y - \sec^2 y)}{\frac{y}{x}\sec y - \tan y}$$

$$= \frac{-1 + \frac{1}{x}\sec y - \frac{1}{x}\sec y - \frac{y}{x}\sec y \tan y + \sec^2 y}{\frac{y}{x}\sec y - \tan y}$$

$$= \frac{-1 - \frac{y}{x}\sec y \tan y + \sec^2 y}{\frac{y}{x}\sec y - \tan y}$$

$$= \frac{-\frac{y}{x}\sec y \tan y + \tan^2 y}{\frac{y}{x}\sec y - \tan y}$$

$$= \frac{(-\frac{y}{x}\sec y + \tan y)\tan}{\frac{y}{x}\sec y - \tan y} = -\tan y = f(y)$$

$$I.F = e^{\int -\tan y dy} = e^{-\log \sec y} = \cos y$$

Multiplying the given equation by the integrating factor, we get

$$(\frac{y}{x}\sec y - \tan y)\cos y dx - (x - \sec y \log x)\cos y dy = 0$$

$$(y/x - \sin x)\, dx - (x \cos y - \log x)\, dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{1}{x} - \cos y + \frac{1}{y}$$

The given Equation is an exact differential Equation.

$$\int_{y=\text{constant}} M dx + \int (\text{Terms in } N \text{ not containing } x) dy = C$$

$$\int (y/x - \sin y) dx - \int 0\, dy = C$$

$$y \log x - x \sin y = C$$

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3. Solve $(3x^2y^4+2xy)\,dx+(2x^3y^3-x^2)\,dy=0$

Soln:- Let $(3x^2y^4+2xy)\,dx+(2x^3y^3-x^2)\,dy=0$ — (1) be the give DE.

Comparing with $Mdx+Ndy=0$, we get

$M=3x^2y^4+2xy$ and $N=2x^3y^3-x^2$

$\frac{\partial M}{\partial y}=3x^2(4y^3)+2x(1)=12x^2y^3+2x$

$\frac{\partial N}{\partial x}=2y^3(3x^2)-(2x)=6x^2y^3-2x$

Here, $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$

The given differential Equation is not exact differential equation.

Consider,

$$\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}=(6x^2y^3-2x)-(12x^2y^3+2x)$$

$$=6x^2y^3-2x-12x^2y^3-2x$$

$$=-6x^2y^3-4x$$

we have,

$$\frac{\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}}{M}=\frac{-6x^2y^3-4x}{3x^2y^4+2xy}=\frac{-2(3x^2y^3+2x)}{y(3x^2y^3+2x)}$$

$$=\frac{-2}{y}=g(y)$$

$$IF=e^{\int g(y)\,dy}=e^{\int\frac{-2}{y}\,dy}$$

$$=e^{-2\int 1/y\,dy}=e^{-2\log y}=e^{\log y^{-2}}$$

$$IF=y^{-2}=\frac{1}{y^2}$$

Multiplying given differential equation by IF, we get

$$\left[(3x^2y^4+2xy)\,dx\right]\frac{1}{y^2}+\left[(2x^3y^3-x^2)dy\right]\frac{1}{y^2}=0$$

$$\left(3x^2y^2+\frac{2x}{y}\right)dx+\left(2x^3y-\frac{x^2}{y^2}\right)dy=0 \quad \text{—— (2)}$$

$M' = 3x^2y^2+\frac{2x}{y}$ and $N' = 2x^3y-\frac{x^2}{y^2}$

$$\frac{\partial M'}{\partial y}=\frac{\partial}{\partial y}\left(3x^2y^2+\frac{2x}{y}\right)=3x^2(2y)+2x\left(-\frac{1}{y^2}\right)$$

$$\therefore \frac{\partial M'}{\partial y}=6x^2y-\frac{2x}{y}$$

$$\frac{\partial N'}{\partial x}=\frac{\partial}{\partial x}\left(2x^3y-\frac{x^2}{y^2}\right)=2y(3x^2)-\frac{1}{y^2}(2x)$$

$$\frac{\partial N'}{\partial x}=6x^2y-\frac{2x}{y^2}$$

Now, $\frac{\partial M'}{\partial y}=\frac{\partial N'}{\partial x}$

∴ The differential Equation (2) is an exact differential equation.

The general solution of Equation (2) is given as,

$$\int_{y=\text{constant}} M'dx+\int(\text{terms of } N' \text{ not containing } x)dy=C$$

$$\therefore \int_{y=\text{constant}}\left(3x^2y^2+\frac{2x}{y}\right)dx+\int 0\,dy=C$$

$$\therefore 3y^2\int x^2dx+\frac{2}{y}\int x\,dx=C$$

$$\therefore y^2x^3+\frac{x^2}{y}=C$$

4. Solve $(2xy^2-y)dx+x\,dy=0$

Soln:- Let $(2xy^2-y)dx+x\,dy=0$ —— (1) be the differential Equation

Comparing with $Mdx+Ndy=0$

$M=2xy^2-y$, $N=x$

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$\therefore \frac{\partial M}{\partial y} = 4xy - 1 \quad , \frac{\partial N}{\partial x} = 1$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\therefore$ The given equation ① is not exact.

$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1 - (4xy - 1) = 2 - 4xy$

$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{2 - 4xy}{2xy^2 - y} = -\frac{2}{y} = f(y)$

$\therefore f(y) = -\frac{2}{y}$

$IF = e^{\int f(y)\,dy} = e^{\int (-2/y)\,dy} = e^{-2\ln y} = \frac{1}{y^2}$

Multiplying given equation by an IF, we get

$\left(2x - \frac{1}{y}\right) dx + \frac{x}{y^2} dy = 0$ — ②

Now,

$M' = 2x - \frac{1}{y} \quad , N' = \frac{x}{y^2}$

$\frac{\partial M'}{\partial y} = \frac{1}{y^2} \quad , \frac{\partial N'}{\partial x} = \frac{1}{y^2}$

$\therefore$ The given differential equation is exact.

$\therefore$ The general solution of equation ② is given as,

$\int_{y = \text{Constant}} M'dx + \int ([\text{Terms of } N \text{ not containing } x)\, dy = C$

$\therefore \int \left(2x - \frac{1}{y}\right) dx + \int (0)\,dy = C$

$\therefore x^2 - \frac{x}{y} = C$

Problems on Rule 3

1. Solve $y(xy+2x^2y^2)dx+x(xy-x^2y^2)dy=0$

Soln:- Let $y(xy+2x^2y^2)dx+x(xy-x^2y^2)dy=0$ —① be the given DE.

Equating the given differential Equation with $Mdx+Ndy=0$, we get

$M = y(xy+2x^2y^2) = xy^2+2x^2y^3$

$N = x(xy-x^2y^2) = x^2y-x^3y^2$

$\frac{\partial M}{\partial y} = 2xy+6x^2y^3$, $\frac{\partial N}{\partial x} = 2xy-3x^2y^2$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ Given differential Equation is not exact.

But the given differential equation is of the form $f_1(xy)ydx+f_2(xy)xdy=0$

$Mx-Ny = y(xy+2x^2y^2)x-x(xy-x^2y^2)y$

$= (xy+2x^2y^2-xy+x^2y^2)xy$

$= (3x^2y^2)xy = 3x^3y^3 \neq 0$

$IF = \frac{1}{Mx-Ny} = \frac{1}{3x^3y^3}$

Multiplying the given differential equation by an IF,

$[y(xy+2x^2y^2)dx]\frac{1}{3x^3y^3}+[x(xy-x^2y^2)dy]\frac{1}{3x^3y^3}=0$

$\left[\frac{xy+2x^2y^2}{3x^3y^2}\right]dx+\left[\frac{xy-x^2y^2}{3x^2y^3}\right]dy=0$

$\left[\frac{1}{3x^2y}+\frac{2}{3x}\right]dx+\left[\frac{1}{3xy^2}-\frac{1}{3y}\right]dy=0$ —①

$M' = \frac{1}{3x^2y}+\frac{2}{3x}$, $N' = \frac{1}{3xy^2}-\frac{1}{3y}$

$\frac{\partial M'}{\partial y} = -\frac{1}{3x^2y^2}$, $\frac{\partial N'}{\partial x} = -\frac{1}{3x^2y^2}$

Here, $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$

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∴ The differential Equation ② is an exact
∴ The general solution of the equation ② is given as,

$$\int_{y=\text{constant}} M'dx + \int [\text{Terms of } N \text{ free from } x]\,dy = C$$

$$\therefore \int \left(\frac{1}{3x^2y^2} + \frac{2}{3x}\right)dx + \int \left(\frac{-1}{3y}\right)dy = C$$

$$\therefore \frac{1}{3y}\int \frac{1}{x^2}\,dx + \frac{2}{3}\int \frac{1}{x}\,dx - \frac{1}{3}\int \frac{1}{y}\,dy = C$$

$$\therefore \frac{1}{3y}\left[\frac{x^{-2+1}}{-2+1}\right] + \frac{2}{3}(\log x) - \frac{1}{3}(\log y) = C$$

$$\therefore -\frac{1}{xy} + 2\log x - \log y = C$$

OR $$-\frac{1}{xy} + \log\frac{x^2}{y} = C$$

2. Solve $y(x^2y^2+2)dx + x(2-2x^2y^2)dy = 0$

Soln:- Let, $y(x^2y^2+2)dx + x(2-2x^2y^2)dy = 0$ — ①
be the given differential equation.
Comparing with $Mdx + Ndy = 0$, we get,

$M = y(x^2y^2+2) = x^2y^3 + 2y$
$N = x(2-2x^2y^2) = 2x - 2x^3y^2$

The given differential equation is of the form, $f_1(xy)y\,dx + f_2(xy)x\,dy = 0$

Consider $Mx - Ny = y(x^2y^2+2)x - x(2-2x^2y^2)y$
$= [x^2y^2 + 2 - 2 + 2x^2y^2]xy$
$= (3x^2y^2)xy = 3x^3y^3 \neq 0$

we have, $IF = \frac{1}{Mx-Ny} = \frac{1}{3x^3y^3}$

Multiplying the given differential equation ① by an IF,

$$[y(x^2y^2+2)dx]\frac{1}{3x^3y^3} + [x(2-2x^2y^2)dy]\frac{1}{3x^3y^3} = 0$$

$$\left(\frac{x^2y^2+2}{3x^3y^2}\right)dx + \left(\frac{2-2x^2y^2}{3x^2y^3}\right)dy = 0$$

$$\left(\frac{1}{3x} + \frac{2}{3x^3y^2}\right)dx + \left(\frac{2}{3x^2y^3} - \frac{2}{3y}\right) = 0 \quad \text{—— ②}$$

$$M' = \frac{1}{3x} + \frac{2}{3x^3y^2}, \quad N' = \frac{2}{3x^2y^3} - \frac{2}{3y}$$

$$\frac{\partial M'}{\partial y} = \frac{-4}{3x^3y^3}, \quad \frac{\partial N'}{\partial x} = \frac{-4}{3x^3y^3}$$

$$\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$$

∴ The differential Equation ② is an exact differential equation.

∴ The general solution of Equation ② is given as,

$$\int M'dx + \int (\text{terms of } N' \text{ not containing } x)\,dy = C$$

(y = constant)

$$\therefore \int \frac{1}{3x} + \frac{2}{3x^3y^2}\,dx + \int\left(\frac{-2}{3y}\right)dy = C$$

$$\frac{1}{3}\int\frac{1}{x}dx + \frac{2}{3y^2}\int\frac{1}{x^3}dx - \frac{2}{3}\int\frac{1}{y}dy = C$$

$$\therefore \frac{1}{3}(\log x) + \frac{2}{3y^2}\left[\frac{x^{-3+1}}{-3+1}\right] - \frac{2}{3}(\log y) = C$$

$$\therefore \log\frac{x}{y^2} - \frac{1}{x^2y^2} = C$$

3. Solve $\frac{dy}{dx} = -\frac{x^2y^3+2y}{2x-2x^3y^2}$

Soln:- Let, $\frac{dy}{dx} = -\frac{x^2y^3+2y}{2x-2x^3y^2}$

Rewriting the given differential equation as $Mdx+Ndy=0$

$$-(x^2y^3+2y)dx = (2x-2x^3y^2)\,dy$$

$$(x^2y^3+2y)dx + (2x-2x^3y^2)\,dy = 0$$

$$y(x^2y^2+2)dx + x(2-2x^2y^2)\,dy = 0$$

Comparing with $Mdx + Ndy = 0$, we get

$M = y(x^2y^2+2)\,dx$, $N = x(2-2x^2y^2)\,dy = 0$

Here,

$f_1(xy) = x^2y^2+2$, $f_2(xy) = 2-2x^2y^2$

$\therefore I.F = \frac{1}{Mx-Ny}$

$Mx = xy(x^2y^2+2)$ $\quad Ny = xy(2-2x^2y^2)$

$Mx - Ny = xy[(x^2y^2+2) - (2-2x^2y^2)]$

$\therefore I.F = \frac{1}{3x^3y^3}$

Multiplying the equation by I.F,

$$\frac{x^2y^3+2y}{3x^3y^3}\,dx + \frac{2x-2x^3y^2}{3x^3y^3}\,dy = 0 \quad -①$$

$$\left(\frac{1}{3x} + \frac{2}{3x^3y^2}\right)dx + \left(\frac{2}{3x^2y^3} - \frac{2}{3y}\right)dy = 0$$

$\therefore$ The given differential equation is exact.

$\therefore$ The general solution of equation ① is given as,

$$\int_{y=\text{constant}} Mdx + \int (N \text{ terms of free from } x)\,dy = C$$

$$\int \frac{1}{3x} + \frac{2}{3x^3y^2}\,dx + \int -\frac{2}{3y}\,dy = C$$

$$\frac{1}{3}\ln x - \frac{1}{3x^2y^2}\,dx + \left(-\frac{2}{3y}\right) = C$$

$$\boxed{\therefore \ln\left(\frac{x}{y^2}\right) - \frac{1}{x^2y^2} = C}$$

4. Solve $(y-xy^2)\,dx - (x+x^2y)\,dy = 0$

Soln:- Let $(y-xy^2)\,dx - (x+x^2y)\,dy = 0$ — ① be the

given differential equation.

Equating the given differential equation with $Mdx+Ndy=0$,

$M = y - xy^2$, $N = -(x+x^2y) = -x - x^2y$

$\frac{\partial M}{\partial y} = 1-2xy$, $\frac{\partial N}{\partial x} = -1-2xy$

Here $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so given differential equation is not an exact equation.

Rewritting the given differential equation,

$(1-xy)y\,dx - (1+xy)x\,dy = 0$

This is of the $f_1(xy)y\,dx + f_2(xy)x\,dy = 0$

$Mx - Ny = 2xy \neq 0$

$\therefore I.F = \frac{1}{Mx-Ny} = \frac{1}{2xy}$

Multiplying differential equation ① by IF, we get

$\frac{1}{2xy}(y-xy^2)dx - \frac{1}{2xy}(x+x^2y)dy = 0$

$\frac{1}{xy}(y-xy^2)dx - \frac{1}{xy}(x+x^2y)dy = 0$

$\left(\frac{1}{x}-y\right) - \left(\frac{1}{y}+x\right)dy = 0$ ②

The differential equation ② is exact DE

$\therefore$ The general solution of Equation ② is given as,

$\int M\,dx + \int [\text{Terms in } N \text{ free from } x]\,dy = C$

$y = \text{constant}$

$\int \left(\frac{1}{x}-y\right) - \int \frac{1}{y}dy = C$

$\log x - xy - \log y = C$

$\therefore \log\left(\frac{x}{y}\right) - xy = C$

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5. Solve $(y(xy - 2x^2y^2)dx + x(xy + x^2y^2)dy = 0$

Soln:- Let $y(xy - 2x^2y^2)dx + x(xy + x^2y^2)dy = 0$ — ①

be the given differential equation.

The above equation ① is of the form

$$f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0$$

Comparing with $Mdx + Ndy = 0$, we get

$$M = y(xy - 2x^2y^2)$$
$$N = x(xy + x^2y^2)$$

Here,

$$M = y f_1(xy), \quad N = x f_2(xy)$$

$$\therefore I.F = \frac{1}{Mx - Ny}$$

$$\therefore I.F = \frac{1}{Mx - Ny} = \frac{1}{xy(xy - 2x^2y^2) - (xy + x^2y^2)}$$

$$\therefore I.F = \frac{1}{x^3y^3}$$

Multiplying equation ① by I.F, we get,

$$\frac{y(xy - 2x^2y^2)}{x^3y^3}dx + \frac{x(xy + x^2y^2)}{x^3y^3}dy = 0$$

$$\left(\frac{1}{x^2y} - \frac{2}{x}\right)dx + \left(\frac{1}{xy^2} + \frac{1}{y}\right)dy = 0 \quad - ②$$

Now, the equation is exact.

The general solution of equation ② is given as,

$$\int M'dx + \int [\text{Terms of } N \text{ free from } x]$$
$$\int 2xy - 6x^2y^2\,dx + \int 0\,dy = C$$

$$\therefore x^2y - 2x^3y^2 = C$$

Problems on Rule 4

1. Solve $(x^2+y^2)dx-(x^2+xy)dy=0$

Soln: Let $(x^2+y^2)\,dx-(x^2+xy)\,dy=0$ be the given DE.

Comparing with $Mdx+Ndy=0$, we get,

$M=x^3+y^2$, $N=-(x^2+xy)=-x^2-xy$

$\frac{\partial M}{\partial y}=2y$, $N=-2x-y$

Here, $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$

The given differential equation is not exact equation, But the given differential equation is homogeneous differential equation.

Consider,

$$Mx+Ny=(x^2+y^2)x+(-x^2-xy)y$$
$$=x^3+xy^2-x^2y-xy^2=x^3-x^2y\neq 0$$

we have $IF=\frac{1}{Mx+Ny}=\frac{1}{x^3-x^2y}$

Multiplying the given differential equation by an IF, we get, $[(x^2+y^2)dx]\frac{1}{x^3-x^2y}-[(x^2+xy)dy]\frac{1}{x^3-x^2y}=0$

$$\left(\frac{x^2+y^2}{x^3-x^2y}\right)dx-\left(\frac{x+y}{x^2-xy}\right)dy=0 \quad -(2)$$

$M'=\frac{x^2+y^2}{x^3-x^2y}$, $N'=-\left(\frac{x+y}{x^2-xy}\right)$

$\frac{\partial M'}{\partial y}=\frac{x^2+2xy-y^2}{x^2(x-y)^2}$, $\frac{\partial N'}{\partial x}=\frac{x^2+2xy-y^2}{x^2(x-y)^2}$

Now, $\frac{\partial M'}{\partial y}=\frac{\partial N'}{\partial x}$

∴ The differential equation (2) is an exact differential equation.

∴ The general solution of Equation (2) is given as

$\int_{y=\text{constant}} M'dx+\int(\text{Terms of } N' \text{ not containing } x)\,dy=c$

$$\int \frac{x^2+y^2}{x^3-x^2y}\,dx + \int (0)\,dy = C$$

$$\therefore \int_{y=\text{constant}} \frac{x^2}{x^2(x-y)}\,dx + y^2 \int_{y=\text{constant}} \frac{1}{x^2(x-y)}\,dx = C$$

Using partial fractions, we get

$$\therefore \int_{y=\text{constant}} \frac{1}{x-y}\,dx + y^2 \int_{y=\text{constant}} \left[\frac{1}{y^2}\frac{1}{x-y} - \frac{1}{y^2}\frac{1}{x} - \frac{1}{y}\frac{1}{x^2}\right]dx = C$$

$$\therefore \int_{y=\text{constant}} \frac{1}{x-y}\,dx + \int \frac{1}{x-y}\,dx - \int \frac{1}{x}\,dx - y\int \frac{1}{x^2}\,dx = C$$

$$\therefore 2\left(\frac{\log(x-y)}{(1-0)}\right) - \log x - y\left[\frac{x^{-2+1}}{-2+1}\right] = C$$

$$\boxed{\therefore\ 2\log(x-y) - \log x + \frac{y}{x} = C}$$

2. Solve $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$

Soln:- Let $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$ —(1) be the given differential equation.

Comparing with $M\,dx + N\,dy = 0$, we get,

$$M = x^2y - 2xy^2$$

$$N = -(x^3 - 3x^2y) = -x^3 + 3x^2y$$

$$\frac{\partial M}{\partial y} = x^2 - 4xy \quad , \quad \frac{\partial N}{\partial x} = -3x^2 + 6xy$$

Here, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

Given differential equation is not an exact equation.

But the given differential equation is a homogenous equation.

Consider,

$$Mx + Ny = (x^2y - 2xy^2)x + (-x^3 + 3x^2y)y$$

$= x^3y - 2x^2y^2 - x^3y + 3x^2y^2 = x^2y^2 \neq 0$

we have $IF = \frac{1}{Mx+Ny} = \frac{1}{x^2y^2}$

Multiplying the given differential equation by an IF, we get

$[(x^2y - 2xy^2)dx]\frac{1}{x^2y^2} - [(x^3 - 3x^2y)dy]\frac{1}{x^2y^2} = 0$

$\left(\frac{1}{y} - \frac{2}{x}\right)dx - \left(\frac{x}{y^2} - \frac{3}{y}\right)dy = 0$ — (2)

$M' = \frac{1}{y} - \frac{2}{x}$

$N' = -\left(\frac{x}{y^2} - \frac{3}{y}\right) = -\frac{x}{y^2} + \frac{3}{y}$

$\frac{\partial M'}{\partial y} = \frac{\partial}{\partial y}\left(\frac{1}{y} - \frac{2}{x}\right) = \frac{\partial(1/y)}{\partial y} - \frac{2}{x}\frac{\partial 1}{\partial y} = \frac{-1}{y^2} - \frac{2}{x}(0) = \frac{-1}{y^2}$

$\frac{\partial N'}{\partial x} = \frac{\partial}{\partial x}\left(\frac{-x}{y^2} + \frac{3}{y}\right) = \frac{-1\,\partial x}{y^2 \partial x} + \frac{3}{y}\frac{\partial 1}{\partial x}$

$= \frac{-1}{y^2}(1) + \frac{3}{y}(0) = \frac{-1}{y^2}$

Now, $\frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$

∴ The differential Equation (2) is an exact differential equation.

∴ The general solution can be given as,

$\int M'dx + \int$(terms of N free from x)$dy = C$ (y=constant)

$\therefore \int\left(\frac{1}{y} - \frac{2}{x}\right)dx + \int\left(\frac{3}{y}\right)dy = C$

$\therefore \frac{1}{y}\int dx - 2\int\frac{1}{x}dx + 3\int\frac{1}{y}dy = C$

$\therefore \frac{1}{y}(x) - 2\log x + 3\log y = C$

$\therefore \frac{x}{y} - 2\log x + 3\log y = C$

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3. Solve $(x^2 - xy + y^2)dx - xydy = 0$

Soln: Let $(x^2 - xy + y^2)dx - xydy = 0$ — ①
be the differential equation.

The above DE is homogeneous differential equation,

we have, $M = x^2 - xy + y^2$ and $N = -xy$

Consider $Mx + Ny = x^3 - x^2y + xy^2 - xy^2$

$Mx + Ny = x^3 - x^2y = x^2(x-y) \neq 0$

we have $IF = \frac{1}{Mx+Ny} = \frac{1}{x^2(x-y)}$

Multiplying Equation ① by IF we get,

$$\frac{1}{x^2(x-y)}(x^2 - xy + y^2)dx - \frac{1}{x^2(x-y)}(xy)dy = 0$$

$$\frac{x^2 - xy}{x^2(x-y)}dx + \frac{y^2}{x^2(x-y)}dx - \frac{xy}{x^2(x-y)}dy = 0$$

$$\frac{1}{x}dx + \frac{y^2}{x^2(x-y)}dx - \frac{y}{x(x-y)}dy = 0 \quad — ②$$

The above equation ② is exact DE.

The general solution of Equation ② is given as,

$$\int_{y=\text{constant}} Mdx + \int (\text{terms in } N' \text{ free from } x)dy = C$$

$$\int \left[\frac{1}{x} + \frac{y^2}{x^2(x-y)}\right]dx = C$$

Using partial fractions, we get

$$\int \left[\frac{1}{x} + \left(\frac{1}{x-y} - \frac{1}{x} - \frac{y}{x^2}\right)\right]dx = C$$

$$\int \left[\frac{1}{x-y} - \frac{y}{x^2}\right]dx = C$$

$$\log(x-y) + \frac{y}{x} = C$$

$$x\log(x-y) + y = Cx$$

4. Solve $(x^4+y^4)\,dx - xy^3\,dy = 0$

Soln: Let $(x^4+y^4)dx - xy^3dy = 0$ —— ① be the given DE.

Comparing with $Mdx + Ndy = 0$, we get,

$M = x^4+y^4$ and $N = -xy^3$

The given Equation ① is homogeneous differential equation.

$$Mx+Ny = (x^4+y^4)x - xy^4 = x^5+xy^4-xy^4 = x^5$$

$$Mx+Ny = x^5 \neq 0$$

we have $IF = \dfrac{1}{Mx+Ny} = \dfrac{1}{x^5}$

Multiplying Equation ① by IF, we get

$$\frac{1}{x^5}(x^4+y^4)\,dx - \frac{1}{x^5}(xy^3)\,dy = 0$$

$$\left(\frac{1}{x} + \frac{y^4}{x^5}\right)dy - \frac{y^3}{x^4}\,dy = 0 \quad —— ②$$

we have

$$M' = \frac{1}{x} + \frac{y^4}{x^5} \quad \text{and} \quad N' = -\frac{y^3}{x^4}$$

$$\frac{\partial M'}{\partial y} = \frac{4y^3}{x^5} \quad \text{and} \quad \frac{\partial N'}{\partial x} = \frac{4y^3}{x^5}$$

∴ The equation ② is exact differential equation.

The general solution of equation ② is given as,

$$\int_{y=\text{constant}} M'dx + \int [\text{terms of } N' \text{ free from } x]\,dy = C$$

$$\int \left(\frac{1}{x} + \frac{y^4}{x^5}\right)dx = C$$

$$\log x + y^4\left(\frac{x^{-4}}{-4}\right) = C$$

$$\log x + \frac{y^4}{-4x^4} = C$$

$$\therefore \log x - \frac{y^4}{4x^4} = C$$

Problems on Linear Differential Equation.

1. Solve $\frac{dy}{dx} + \frac{2x}{x^2+1} y = \frac{4x}{x^2+1}$

Soln: Let $\frac{dy}{dx} + \frac{2x}{x^2+1} y = \frac{4x}{x^2+1}$ be the given LDE in y.

Comparing the given LDE with $\frac{dy}{dx} + Py = Q$, we get,

$P = \frac{2x}{x^2+1}$, $Q = \frac{4}{x^2+1}$

Consider, $\int P dx = \int \frac{2x}{x^2+1} dx = \ln(x^2+1)$

$$y(x^2+1) = \int (x^2+1) \frac{4x}{x^2+1} dx + C$$

$$y(x^2+1) = \int 4x dx = 2x^2 + C$$

$$\therefore y = \frac{2x^2+C}{x^2+1}$$

2. Solve $\sin 2x \frac{dy}{dx} = y + \tan x$

Soln:- Let $\sin 2x \cdot \frac{dy}{dx} = y + \tan x$ be the given DE.

$$\sin 2x \cdot \frac{dy}{dx} - y = \tan y$$

$$\frac{dy}{dx} - \frac{y}{\sin 2x} = \frac{\tan x}{\sin 2x} \quad \text{---} \quad (1)$$

$\left\{ \frac{dy}{dx} + Py = 0 \right\}$

The above DE is LDE in y.

we have $P = \frac{-1}{\sin 2x}$ and $Q = \frac{\tan x}{\sin 2x} = \frac{\sin x}{2 \sin x \cdot \cos x \cos x}$

$Q = \frac{1}{2}\sec^2 x$

Let $\int P dx = \int \frac{-1}{\sin 2x} dx = \int -\operatorname{cosec} 2x \, dx$

$= -\int \operatorname{cosec} 2x = -\log(\operatorname{cosec} 2x - \cot 2x)$

$\int P dx = \frac{-1}{2}\log(\operatorname{cosec} 2x - \cot 2x)^2$

$\int P dx = \log(\operatorname{cosec} 2x - \cot 2x)^{-1/2}$

$e^{\int P dx} = e^{\log(\operatorname{cosec} 2x - \cot 2x)^{-1/2}}$

$= (\operatorname{cosec} 2x - \cot 2x)^{-1/2}$

$= \left[\frac{1}{\sin 2x} - \frac{\cos 2x}{\sin 2x}\right]^{-1/2}$

$= \left[\frac{1-\cos 2x}{\sin 2x}\right]^{-1/2}$

$= \frac{-1}{2}\log(\tan x) = \log\left(\frac{1}{\sqrt{\tan x}}\right)$

$\int P dx = \log\left(\frac{1}{\sqrt{\tan x}}\right)$

$e^{\int P dx} = e^{\log\left(\frac{1}{\sqrt{\tan x}}\right)} - \frac{1}{\sqrt{\tan x}}$

The general solution of Equation ① is given as,

$y e^{\int P dx} = \int Q \cdot e^{\int P dx} dx + C$

$y \frac{1}{\sqrt{\tan x}} = \int \frac{1}{2}\sec^2 x \cdot \frac{1}{\sqrt{\tan x}} dx + C$

$\frac{y}{\sqrt{\tan x}} = \int \frac{1}{2\sqrt{\tan x}} \cdot \sec^2 x \, dx + C = \sqrt{\tan x} + C$

$\boxed{y = \tan x + C \cdot \sqrt{\tan x}}$

3. Solve $\cos x \frac{dy}{dx} + y \sin x = \sec x$

Soln:- Let $\cos x \frac{dy}{dx} + y \sin x = \sec$ be the given DE.

$\frac{dy}{dx} + y \tan x = \sec^2 x$

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which is LDE with,

$P = \tan x$, $Q = \sec^2 x$

$\therefore e^{\int P dx} = e^{\log(\sec x)} = \sec x$

$y \sec x = \int \sec x \cdot \sec^2 x \, dx + C$

$= \int \sec^3 x \, dx$

$y \sec x = \frac{1}{2} (\sec x \cdot \tan x + \log \sec x + \tan x) + C$

$y = \frac{1}{2} (\tan x + \sec x \log \sec x + \tan x) + C \sec x$

4. Solve $(1-x^2)\frac{dy}{dx} + 2xy = x\sqrt{1-x^2}$

Soln:- Let $(1-x^2)\frac{dy}{dx} + 2xy = x\sqrt{1-x^2}$ be the give DE.

$$\frac{dy}{dx} + \frac{2x}{1-x^2} y = \frac{x}{\sqrt{1-x^2}}$$

Now consider,

$$P = \frac{2x}{1-x^2} \quad , \quad Q = \frac{x}{\sqrt{1-x^2}}$$

$$\int P dx = -\log(1-x^2)$$

$$e^{\int P dx} = \frac{1}{1-x^2}$$

$$\frac{y}{1-x^2} = \int \frac{x}{(1-x^2)^{3/2}} dx + C$$

$$y = \frac{1}{\sqrt{1-x^2}} + C$$

$$\therefore y = \sqrt{1-x^2} + C(1-x^2)$$

5. Solve $(1+y^2)\,dx = (e^{\tan^{-1}y} - x)\,dy$

Soln:- Let $(1+y^2)dx = (e^{\tan^{-1}y} - x)\,dy$ be the given DE.

$$\frac{dx}{dy} = \frac{e^{\tan^{-1}y} - x}{1+y^2} = \frac{e^{\tan^{-1}y}}{1+y^2} - \frac{x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{1}{1+y^2}x = \frac{e^{\tan^{-1}y}}{1+y^2} \quad \text{---} \quad (1)$$

This equation is LDE in x.

Comparing the given LDE with $\frac{dx}{dy} + P'x = Q'$

we get $P' = \frac{1}{1+y^2}$ and $Q' = \frac{e^{\tan^{-1}y}}{1+y^2}$

Consider $e^{\int P'dy} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$

The solution of given LDE (1) is given as

$$x e^{\int P'dy} = \int e^{\int P'dy} Q'\,dy + C$$

$$\therefore x(e^{\tan^{-1}y}) = \int e^{\tan^{-1}y} \times \frac{e^{\tan^{-1}y}}{1+y^2}\,dy + C$$

$$\therefore x e^{\tan^{-1}y} = \frac{[e^{\tan^{-1}y}]^2}{2} + C$$

[Because $\int f(x)\,f'(x)\,dx = \frac{[f(x)]^2}{2}$ and taking $f(x) = e^{\tan^{-1}y}$]

$$\therefore x = \frac{1}{2}e^{\tan^{-1}y} + \frac{c}{e^{\tan^{-1}y}}$$

Problems on reducible to Linear Differential Equation

1. Solve $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$

Soln: Let $\frac{dy}{dx} + x\cdot\sin y = x^3\cos^2 y$ be the given DE.

$$\frac{1}{\cos^2 x}\cdot\frac{dy}{dx} + \frac{\sin^2 y}{\cos^2 y}\cdot x = x^3$$

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$$\sec^2 y \frac{dy}{dx} + \frac{2\sin y \cdot \cos y}{\cos^2 y} \cdot x = x^3$$

$$\sec^2 y \frac{dy}{dx} + 2x \cdot \tan y = x^3 \quad \text{— (1)}$$

$$\left\{ f'(y) \frac{dy}{dx} + P f(y) = Q \right\}$$

Put $\tan y = V$ $\quad \sec^2 y \frac{dy}{dx} = \frac{dV}{dx}$

Equation (1) becomes

$$\frac{dV}{dx} + 2x \cdot V = x^3 \quad \text{— (2)}$$

which is LDE in V with

$P = 2x$ and $Q = x^3$

$\int P dx = \int 2x\,dx = x^2$

The general solution of eqn (1) is given as

$$V e^{\int P dx} = \int e^{\int P dt} \cdot Q\,dx + C$$

$$V e^{x^2} = \int e^{x^2} \cdot x^3 \cdot dx + C$$

$$= \int e^{x^2} \cdot x^2 \cdot x\,dx + C$$

$\therefore$ Put $x^2 = t$ $\quad 2x\,dx = dt$ $\quad \therefore x\,dx = \frac{dt}{2}$

we get

$$V e^{x^2} = \int e^t \cdot t \cdot \frac{dt}{2} + C$$

$$= \frac{1}{2} \int t \cdot e^t\,dt + C$$

$$= \frac{1}{2} \left[(t-1) e^t \right] + C$$

$$= \frac{1}{2} \left[(t) \cdot e^t - (1) e^t \right] + C$$

Put $t = x^2$, we get

$ve^{v^2} = \frac{1}{2}(x^2-1)e^{x^2} + C$

Put $v = \tan y$, we get

$$\tan y \cdot e^{v^2} = \frac{1}{2}(x^2-1)e^{x^2} + C$$

2. Solve $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

Soln:- Let $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$ be the given DE.

$$\frac{dy}{dx} = e^x \cdot e^{-y}(e^x - e^y)$$

$$\frac{dy}{dx} = \frac{e^{2x}}{e^y} - e^x$$

$$e^y \frac{dy}{dx} = e^{2y} - e^y \cdot e^x$$

$$e^y \frac{dy}{dx} - e^x \cdot e^y = e^{2x} \quad \text{— (1)}$$

$$\left\{ f'(y)\frac{dy}{dx} + Pf(y) = Q \right\}$$

Put $v = e^y$ $\frac{dv}{dx} = e^y \frac{dy}{dx}$, we get

$$\frac{dv}{dx} + e^x \cdot v = e^{2x} \quad \text{— (11)}$$

which is LDE in v with

$P = e^x$ and $Q = e^{2x}$

$\therefore \int P dx = \int e^x dx = e^x$

$\therefore$ The general solution of eqn (2) given as

$$ve^{\int P dx} = \int e^{\int P dx} Q\, dx + C$$

$$ve^{e^x} = \int e^{e^x} \cdot e^{2x} \cdot dx + C$$

$$ve^{e^x} = \int e^{e^x} \cdot e^x \cdot e^x \, dx$$

Put $e^x = t$ $e^x dx = dt$, we get

$$ve^{e^x} = \int e^t \cdot t \cdot dt + C$$

$$ve^{e^x} = (t-1)e^t + C$$

Put $t = e^x$, we get

$ve^{e^x} = (e^x - 1)e^{e^x} + C$

Put $v = e^y$, we get

$$\boxed{e^y e^{e^x} = (e^x - 1)e^{e^x} + C}$$

3 Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$

Soln:- Let $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$ be the given DE

$$\cos y \frac{dy}{dx} - \frac{\tan y \cos y}{1+x} = (1+x)e^x$$

$$\cos y \frac{dy}{dx} - \frac{\sin x}{1+x} = (1+x)e^x$$

Put $v = \sin y$, $\frac{dv}{dx} = \cos y \frac{dy}{dx}$

$$\frac{dv}{dx} - \frac{v}{1+x} = (1+x)e^x$$

$\therefore e^{\int P(x)dx}$, where $P(x) = -\frac{1}{1+x}$

$\therefore IF = e^{\int -\frac{1}{1+x} dx} = e^{-\log(1+x)} = e^{\log(1+x)^{-1}} = \frac{1}{1+x}$

$$\frac{1}{1+x}\left(\frac{dv}{dx} - \frac{v}{1+x}\right) = \frac{1}{1+x}(1+x)e^x$$

$$\frac{d}{dx}\left(\frac{v}{1+x}\right) = e^x$$

$$\frac{v}{1+x} = \int e^x dx = e^x + C$$

Substitute $v = \sin y$, $\frac{\sin y}{1+x} = e^x + C$

$\therefore \sin y = (1+x)\{e^x + C$

4. Solve $3y^2 \frac{dy}{dx} + 2y^3 x = 4x^3 e^{x^2}$

Soln:- Let $3y^2 \frac{dy}{dx} + 2y^3 x = 4x^3 e^{x^2}$ be the given DE.

$\because \frac{d}{dx}(y^3) = 3y^2 \frac{dy}{dx}$

Rewritting the equation,

$$\frac{d}{dx}(y^3) + 2xy^3 = 4x^3 e^{x^2}$$

Now comparing with $\frac{dz}{dx} + Pz = Q$, we get,

$z = y^3$, $P = 2x$, $Q = 4x^3 e^{x^2}$

$I.F = e^{\int 2x\,dx} = e^{x^2}$

$$\frac{d}{dx}(y^3 e^{x^2}) = 4x^3 e^{2x^2}$$

$$y^3 e^{x^2} = \int 4x^3 e^{2x^2} dx$$

Consider, $t = x^2$ then:

$$y^3 e^{x^2} = e^{2x^2}(x^2 - 1) + C$$

$\therefore y^3 e^{x^2} = e^{2x^2}(x^2 - 1) + C$

Problems on Bernoulli's Linear Differential Equation

1. Solve $x \frac{dy}{dx} + y = x^3 y^6$

Soln:- Let $x \frac{dy}{dx} + y = x^3 y^6$ be the given DE.

$$\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$$

$$\frac{1}{y^6}\frac{dy}{dx} + \frac{1}{y^5}\cdot\frac{1}{x} = x^2 \quad \text{---} \quad (1)$$

Put $v = \frac{1}{y^5}$ $\frac{dv}{dx} = \frac{-5}{y^6}\frac{dy}{dx}$

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$$\frac{-1}{5}\frac{dv}{dx} = \frac{1}{y^6}\frac{dy}{dx}$$

we get, $\frac{-1}{5}\frac{dv}{dx} + \frac{1}{x}v = x^2$

$$\frac{dv}{dx} - \frac{5}{x}.v = -5x^2$$

which is LDE in v with $P = \frac{-5}{x}$ and $Q = -5x^2$

$$\int P dx = \int \frac{-5}{x} dx = \log\left(\frac{1}{x^5}\right)$$

$$e^{\int P dx} = e^{\log(1/x^5)} = \frac{1}{x^5}$$

The general solution of eqn (2) is given as

$$v e^{\int P dx} = \int e^{\int P dx} . Q dx + C$$

$$\frac{v}{x^5} = \int \frac{1}{x^5}(-5x^2)dx + C = -5\int \frac{1}{x^3} dx + C = -5\left(\frac{-1}{2x^2}\right) + C$$

$$\frac{v}{x^5} = \frac{5}{2x^2} + C$$

Put $v = \frac{1}{y^5}$, we get

$$\boxed{\frac{1}{x^5y^5} = \frac{5}{2x^2} + C}$$

2. Solve $\frac{dz}{dx} + \frac{z}{x}\log z = \frac{z}{x^2}(\log z)^2$

Soln:- $\frac{1}{z}\frac{dz}{dx} + \frac{1}{x}\log z = \frac{1}{x^2}(\log z)^2$

Let,

$$y = \log z \Rightarrow \frac{dy}{dz} = \frac{1}{z}\frac{dz}{dx}$$

Substitute:

$$\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x^2}y^2$$

$n=2$

$v = y^{1-n} = y^{-1} \Rightarrow \frac{dv}{dx} = -y^{-2}\frac{dy}{dx}$

Multiplying the equation by y^{-2},

$$-\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x^2}$$

$$\frac{dv}{dx} - \frac{1}{x} = -\frac{1}{x^2}$$

$$\therefore I.F = e^{\int -\frac{1}{x}dx} = \frac{1}{x}$$

$$\frac{d}{dx}\left(\frac{v}{x}\right) = \frac{-1}{x^3}$$

$$\frac{v}{x} = \frac{1}{2x^2} + C$$

Substitute $v = \frac{1}{y} = \frac{1}{\log z}$

$$\boxed{\frac{1}{x\log z} = \frac{1}{2x^2} + C}$$

Q.3 Solve $\frac{dr}{d\theta} = r\tan\theta - \frac{r^2}{\cos\theta}$

Soln: Let $\frac{dr}{d\theta} = r\tan\theta - \frac{r^2}{\cos\theta}$ be the given DE.

$$\frac{dr}{d\theta} - r\tan\theta = -r^2\sec\theta$$

This is Bernoulli's LDE in r. Dividing both the sides by r^2, we get,

$$r^{-2}\frac{dr}{d\theta} - \tan\theta\, r^{-1} = -\sec\theta \quad \text{—— (1)}$$

Put $r^{-1} = t \Rightarrow -1r^2\frac{dr}{d\theta} = \frac{dt}{d\theta} \Rightarrow r^{-2}\frac{dr}{d\theta} = -\frac{dt}{d\theta}$

on substituting in Equation ①,

$$\frac{-dt}{d\theta} - \tan\theta\, t = -\sec\theta \quad \frac{dt}{d\theta} + \tan\theta\, t = \sec\theta - ②$$

This is LDE in t with $P' = \tan\theta$ and $Q' = \sec\theta$, we have

$$e^{\int P'd\theta} = e^{\int \tan\theta\, d\theta} = e^{\log\sec\theta} = \sec\theta$$

The general solution of LDE ② is given as,

$$te^{\int P'd\theta} = \int e^{\int P'd\theta} Q'\, d\theta + C$$

$$\therefore t(\sec\theta) = \int \sec\theta \times \sec\theta\, d\theta + C$$

$$\therefore t\sec\theta = \int \sec^2\theta\, d\theta + C$$

$$\therefore t\sec\theta = \tan\theta + C$$

$$\therefore t = \frac{\tan\theta + C}{\sec\theta}$$

Put $t = r^{-1}$, we get

$$\therefore r^{-1} = \sin\theta + C\cos\theta$$

4. Solve $\frac{dy}{dx} + \frac{2}{x}y = \frac{y^3}{x^3}$

Soln: Let $\frac{dy}{dx} + \frac{2}{x}y = \frac{y^3}{x^3}$ be the given DE

Bernoulli with $n = 3$

$$v = y^{-2} \Rightarrow \frac{dv}{dx} = -2y^3 \frac{dy}{dx}$$

Multiplying by -2:

$$\frac{dv}{dx} - \frac{4}{x}v = \frac{-2}{x^3}$$

$$\text{I.F} = e^{\int -\frac{4}{x}dx} = x^{-4}$$

$$\frac{d}{dx}(vx^{-4}) = -2x^{-7}$$

$$\int vx^{-4} = \frac{1}{3}x^{-6} + C$$

Substitute $v = y^{-2}$

$$\boxed{y^{-2} = \frac{1}{3x^2} + cx^4}$$

5. Solve $\frac{dy}{dx} = x^3y^3 - xy$

Soln:- Let, $\frac{dy}{dx} = x^3y^3 - xy$ be the given DE.

$$\frac{dy}{dx} + xy = x^3y^3$$

Bernoulli with $n=3$.

$v = y^{-2} \Rightarrow \frac{dv}{dx} = -2y^{-3}\frac{dy}{dx}$

Multiplying by y^{-3},

$$-\frac{1}{2}\frac{dv}{dx} + xv = x^3$$

Multiplying by -2,

$$\frac{dv}{dx} - 2xv = -2x^3$$

$I.F = e^{\int -2x\,dx} = e^{-x^2}$

$$\frac{d}{dx}(ve^{-x^2}) = -2x^3e^{-x^2}$$

$$\int ve^{-x^2} = e^{-x^2}(x^2+1) + C$$

Substituting $v = y^{-2}$

$$\boxed{y^{-2} = x^2 + 1 + ce^{x^2}}$$

6. Solve $\frac{dx}{dy} = x^3y^3 - xy$

Soln:- Let, $\frac{dx}{dy} = x^3y^3 - xy$ be the given DE.

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$$\frac{dx}{dy} + xy = x^3y^3$$

Bernoulli with $n = 3$

Substitute, $v = x^{-2} \Rightarrow \frac{dv}{dy} = -2x^{-3}\frac{dx}{dy}$

Multiplying by x^{-3},

$$-\frac{1}{2}\frac{dv}{dy} + yv = y^3$$

Multiplying by -2;

$$\frac{dv}{dy} - 2yv = -2y^3$$

$$I.F = e^{\int -2y\,dy} = e^{-y^2}$$

$$\frac{d}{dy}(ve^{-y^2}) = -2y^3e^{-y^2}$$

$$ve^{-y^2} = e^{-y^2}(y^2+1) + C$$

Substitute $v = x^{-2}$,

$$\boxed{x^{-2} = y^2 + 1 + Ce^{y^2}}$$